

Mahak Saleem

Aerospace Engineering Student

Contact Information

Email: Mahak.Saleem@citystgeorges.ac.uk

Location: London, UK

Phone Number: +44 7394 998627

Aerospace Engineering student with experience in **precision manual machining, CAD modelling (Siemens NX), and Python-based numerical analysis**. Demonstrated **leadership** as Lead Manufacturer on a **high-efficiency turbine project**, combining practical workshop execution with theoretical data.



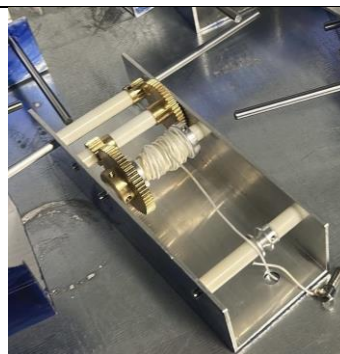
Gear Box Model



Wind Turbine Complete



Gear Box Completed



Internal Mesh Detail

Wind Turbine Project (Gear Box Lead Manufacturer)

Objectives:

- Design and manufacture a **minimum 1:10 overall gear ratio** either using 4 or 6 different combinations of gears.
- Work collaboratively within a group to ensure our 3-blade wind turbine worked efficiently.
- Maintain strict geometric tolerances across gears via **manual machining operations** (lathe/milling) to guarantee **precise shaft alignment and mesh engagement**. Ultimately, ensuring specification criteria met.

Contributions:

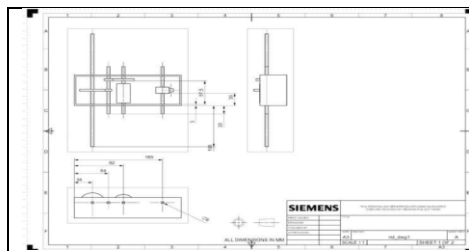
- Calculated **theoretical power efficiency** taking practical aspects into consideration such as frictional losses using Excel.
- **Modelled 3D CAD assemblies (Siemens NX)** and generated annotated engineering drawings verifying critical dimensions ensuring specification criteria met.

Calculation for Shaft A		Calculation for Shaft B	
THIS IS COMPLETED FOR YOU		(to be completed by you)	
omega_A	99.5 rad/s	omega_B	24.9 rad/s
F_g1	5.9 N	F_g2	5.31 N
P_loss,rad	0.52 W	F_g3	15.81 N
P_loss,ax	0.40 W	P_loss,rad	0.47 W
P_loss,tot	0.92 W	P_B,in	3.15 W
P_A,in	4.4 W	P_B,out	2.67 W
P_A,out	3.5 W		

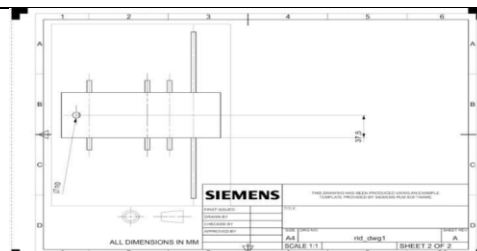
Calculation for Shaft C		Calculation for idler pulley on 'Shaft out'	
(to be completed by you)		(to be completed by you)	
Note that if only 4 gears are used,		omega_out	12.1 rad/s
omega_C	9.9 rad/s	T_string	24.5 N
F_g4	14.2 N	R_out	34.7 N
T_string	24.5 N	P_loss,rad	0.38 W
theta	46.5 degrees	P_out	1.79 W
F_g4'	14.1 N		
error	0.1 N		
R_C	26.3 N		
P_loss,rad	0.23 W		
P_C,in	2.41 W		
P_C,out	2.17 W		

Overall efficiency of the gearbox (= P_out/P_A,in)	
(to be completed by you)	
eta_ov	41% -

Overall Efficiency (Theoretical)



Siemens NX Assembly Specification



Shaft Alignment Orthographic View

Outcomes:

- Achieved overall **module grade of 96.9%.**
- **Achieved 93% mechanical speed efficiency**
- **Demonstrated physical rotation and power transfer** during turbine testing.
- **Secured an 80% mark** on the engineering CAD and technical drawing assessment.